

Scatter plots

A scatter plot places one point for each observed pair of numerical measurements.

What can its overall shape reveal about a possible linear model?

Choosing axes

A shop records advertising spending and weekly sales. It wants to predict sales from spending.

Which variable belongs on the horizontal axis?

Sketching data

Plot these observed pairs: $(1, 4)$, $(2, 6)$, $(3, 7)$, $(4, 9)$, and $(5, 11)$.

Does a line appear to be a reasonable model?

Direction of association

Observed pairs are $(1, 12)$, $(2, 10)$, $(3, 9)$, $(4, 7)$, and $(5, 5)$.

Would a fitted line have positive or negative slope?

A nonlinear pattern

Observed pairs are $(0, 1)$, $(1, 2)$, $(2, 5)$, $(3, 10)$, and $(4, 17)$.

What feature makes a straight line a questionable model?

Estimating a line

A scatter plot trends near $(0, 3)$ and $(6, 15)$.

What line through these two representative points could approximate the trend?

Residuals

A residual is the observed output minus the model prediction.

$$y - \hat{y}$$

What does a positive residual mean on a scatter plot?

Calculating a residual

An observation is (4, 12). The fitted model is:

$$\hat{y} = 2x + 3$$

What is the residual?

Least-squares fitting

A least-squares regression line minimizes the sum of squared residuals.

Why measure vertical differences between observations and predictions?

Using a graphing utility

Enter the pairs $(1, 4)$, $(2, 6)$, $(3, 7)$, $(4, 9)$, and $(5, 11)$.

What least-squares linear model does your graphing utility report?

Interpreting a fitted slope

A model predicts sales y in dollars from advertising x in dollars.

$$\hat{y} = 2.4x + 180$$

What does the slope mean in this model?

Interpolation

A model was fitted using inputs from 10 through 30.

$$\hat{y} = 1.5x + 8$$

What does it predict at $x = 22$?

Extrapolation

A model was fitted using inputs from 10 through 30.

$$\hat{y} = 1.5x + 8$$

What additional concern arises when predicting at $x = 100$?

Correlation

The correlation coefficient r describes the direction and strength of a linear association.

$$-1 \leq r \leq 1$$

Which suggests a stronger linear association: $r = -0.92$ or $r = 0.35$?

A fitted relationship

A fitted line relates advertising spending to sales in observational data.

Does a strong linear association prove that increased spending caused higher sales?